

FUNCTIONAL FEATURES GUIDE

Developer: [Left Blank] | OptiCrop Project Documentation
Repository: <https://github.com/goutham-05-raj/OptiCrop-Smart-Agricultural-Production-Optimization-Engine>

1. Core OptiCrop User Features

- Form Input Validation: Prevents text inputs or negative values in fields like pH and rainfall.
 - Crop Recommendation Display: Renders the recommended crop and system prediction confidence.
 - Yield Forecasting: Displays expected yield in tons per hectare based on seasonal and regional parameters.
 - NPK Diagnostic Prescripts: Highlights nutrient levels and provides recommendations for fertilizers like Urea, SSP, and Potash.
 - Visual Analytics: Displays performance charts (confusion matrices, feature rankings) dynamically.
- Opticrop - Functional Features (5 Implemented)**
1. Crop Recommender
 - 1. Crop Recommender
 - RF classifier recommends optimal crop from 22 types based on N, P, K, pH, rainfall, temperature, humidity.
 2. Yield Forecaster
 - RF regressor estimates harvest yield in t/ha.
 3. Soil Health Diagnoser
 - RF regressor estimates harvest yield in t/ha.
 - K-Means (k=5) clusters soil into Excellent / Good / Moderate / Poor / Critical.
 4. NPK Prescriber
 - K-Means (k=5) clusters soil into Excellent / Good / Moderate / Poor / Critical.
 - Computes NPK deficit vs. crop optimum; recommends Urea (N), SSP (P), MOP (K) kg/ha dosage.
 5. Analytics Dashboard
 - Accuracy comparison bar chart, confusion matrix heatmap, feature-importance plot rendered with Matplotlib/Seaborn.